

A typed discrete–continuous bridge for relaxation labeling, gauge structure and focused representations

Michael Emanuel Glover

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Abstract

Relaxation labeling is usually formulated as inference over a fixed set of labels. We develop a broader, testable formulation in which the object being described and the label used to describe it are roles in one evolving reference-frame state. Attention controls focus; an envelope operator moves attended external evidence to the frame boundary; compatibility combines relational grammar, empirical models and accessible data; and hysteresis permits object–label reversals without chatter. We connect this model to a typed discrete–continuous bridge for triadic gravitational and electromagnetic descriptors. A finite theorem shows that bridge structure, rather than descriptive slot count, controls compatible modes. A triangulated compact- $U(1)$ prototype verifies gauge invariance to 8.9×10^{-16} and converges under refinement. Earlier audits establish exact sparse signed relaxation dynamics, orientation-sensitive chirality, and causal forecasting safeguards. Synthetic phase calculations reproduce interference but include an equivalent paired-real control. The combined evidence supports a coherent computational framework and reproducible software artifacts; it does not yet derive quantum mechanics, general relativity, or a new physical law. We identify the non-flat, matter-coupled, held-out experiment required to test those stronger claims. **Keywords:** relaxation labeling; triadic unity; discrete–continuous bridge; gauge invariance; attention; representation; reproducibility

1 Introduction

Relaxation labeling treats recognition as compatibility-constrained inference: competing hypotheses exchange support until a stable assignment emerges [1, 2]. The framework becomes more expressive when labels are not assumed to exist independently of attention. A feature may remain unresolved object content, become a label when sufficiently focused, or return to object status when focus and evidence decline.

The present work consolidates four formal notes and the accompanying software audits into one research framework. The central construction has two layers. First, a typed bridge relates discrete mesh variables to continuous fields without identifying descriptive roles with physical degrees of freedom. Second, a focused relaxation layer controls which environmental information is admissible to the current reference frame. These layers share compatibility but retain distinct semantics.

The framework is deliberately falsifiable. A theorem is separated from a synthetic verification, and both are separated from a physical conjecture. This distinction matters because phase interference, a zero residual, or a successful finite-element identity can be reproduced by a model whose desired conclusion was inserted at definition time.

2 A focused object–label model

For entity i , define

$$z_i = (r_i, f_i, I_i), \quad r_i \in \{\text{object}, \text{label}\}, \quad f_i \in [0, 1], \quad I_i \geq 0 \quad (1)$$

The role r_i is a state variable rather than an ontology. Focus f_i measures current attention; I_i measures informational richness available for promotion.

For reference frame q , partition evidence into internal E_q^{int} , external E_q^{ext} and boundary B_q . Given attention $a_q(x)$, the envelope is

$$B_q = \text{Env}_\tau(E_q^{\text{ext}}, a_q) = \{x : a_q(x) \geq \tau\}, \quad E_q^{\text{acc}} = E_q^{\text{int}} \cup B_q. \quad (2)$$

The envelope is an accessibility operator: it changes what the model can use, not what exists outside it.

Let G , M and D be support from grammar, empirical models and accessible data. Their provenance-preserving compatibility field is

$$C = \frac{w_G G + w_M M + w_D D}{w_G + w_M + w_D}. \quad (3)$$

The full GRRLE recurrent update can replace this prototype support field without changing the semantics.

Object–label transitions use hysteresis:

$$\text{object} \rightarrow \text{label} \text{ if } f_i \geq \tau_+ \wedge I_i \geq I_{\min}; \quad \text{label} \rightarrow \text{object} \text{ if } f_i \leq \tau_-, \quad (4)$$

with $0 \leq \tau_- < \tau_+ \leq 1$. The gap prevents rapid classification chatter.

The Master Observer is modelled as a read-only map

$$\mathcal{O}(z, E_q^{\text{int}}, B_q, E_q^{\text{ext}}, C) \mapsto (r, f, B_q, C). \quad (5)$$

If observation changes attention, that feedback must be an explicit dynamical operator. This prevents the observer from becoming an unexamined source of causal intervention.

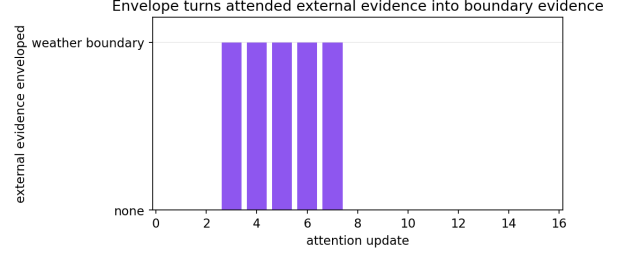
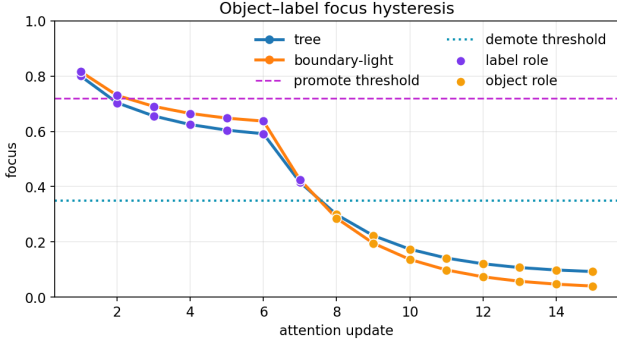


Figure 1: Focused representations. **a**, Focus trajectories with promotion and demotion thresholds; marker colour identifies the current role. **b**, The envelope makes attended external evidence available at the frame boundary.

3 Typed bridge and closure

The twelvefold grammar is

$$\mathcal{I} = \{1, 2, 3\} \times \{G, E\} \times \{D, C\}, \quad |\mathcal{I}| = 12, \quad (6)$$

where the factors index triadic position, physics sector and representation mode. The state is typed:

$$\mathcal{X} = \mathcal{X}_{GD} \times \mathcal{X}_{ED} \times \mathcal{X}_{GC} \times \mathcal{X}_{EC}, \quad (7)$$

not automatically $\mathbb{R}^{12}$ or $\mathbb{C}^{12}$. A role may contain a tensor, a connection, a cochain, a field or a derived observable.

For mesh scale h , restriction and reconstruction are $R_h : \mathcal{X}_C \rightarrow \mathcal{X}_{D,h}$ and $W_h : \mathcal{X}_{D,h} \rightarrow \mathcal{X}_C$. Approximation may satisfy $R_h W_h = I$ on a discrete range while $W_h R_h = \Pi_h \neq I$ on the continuum space. This is representation loss, not a probability law.

Closure is a typed residual:

$$\mathcal{F}_h(X) = (R_G^D, R_E^D, \Pi_h R_G^C, \Pi_h R_E^C, B_G, B_E, B_S) = 0. \quad (8)$$

Thus the structural expression “ $3 + 3 = 0$ ” means that retained sectors have zero coupled residual; it does not mean that the fields disappear.

3.1 Bridge-controlled modes

For $u, v \in \mathbb{R}^6$ and permutation matrix P , define

$$F_P(u, v) = \begin{pmatrix} I & I \\ P & I \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}. \quad (9)$$

Theorem. $\ker F_P = \{(u, -u) : Pu = u\}$ and its dimension equals the number of cycles of P . The proof follows by substituting $v = -u$ and observing that permutation-fixed vectors are constant on cycles. Identity, one transposition and a six-cycle therefore have nullities 6, 5 and 1.

This is a classical constraint theorem. It does not identify compatible modes with particles or quantum states.

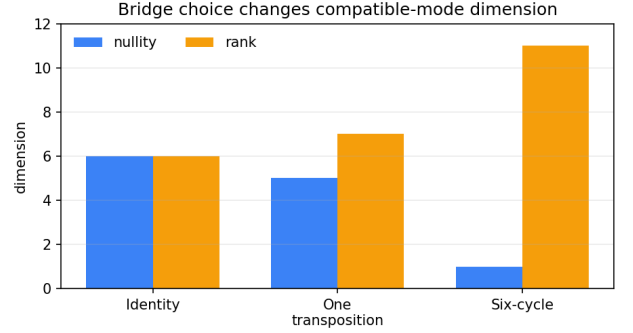


Figure 2: Bridge structure controls compatible-mode dimension. The descriptive slot count is fixed.

4 Gauge and refinement prototype

We triangulate a unit square into counter-clockwise right triangles. The two-dimensional Regge deficit is

$$\epsilon_v = 2\pi - \sum_{t \ni v} \theta_{t,v}. \quad (10)$$

Flat Euclidean edge lengths give $\epsilon_v = 0$. On each oriented edge place a compact gauge link a_e and define triangle flux $\phi = d_1 a$. Under $a \mapsto a + d_0 \chi$, flux is invariant because $d_1 d_0 = 0$.

The prototype action is

$$S_h = \beta_G \sum_v \epsilon_v^2 + \beta_E \sum_t (1 - \cos \phi_t) + \lambda C_h(\epsilon, \phi). \quad (11)$$

The current refinement test uses a smooth constant field $A = (-By/2, Bx/2)$, exact line integration, $C_h = 0$ and $\beta_E = n^2$. Its continuum target is $B^2/4 = 0.25$.

The Wilson action is 0.24967465 on a 2×2 mesh, 0.24999873 on 8×8 , 0.24999992 on 16×16 and 0.25000000 on 32×32 . Gauge-transformed triangle fluxes differ by at most 8.9×10^{-16} . The geometry is flat by construction; this is a gauge and discretization test, not a non-flat Einstein–Maxwell simulation.

5 Representation controls

The history bridge uses $Q_\kappa(h) = \exp(iS[h]/\kappa)$ and $\psi_b = \sum_{h \rightarrow b} w_h Q_\kappa(h)$. A Born-form readout is explicitly

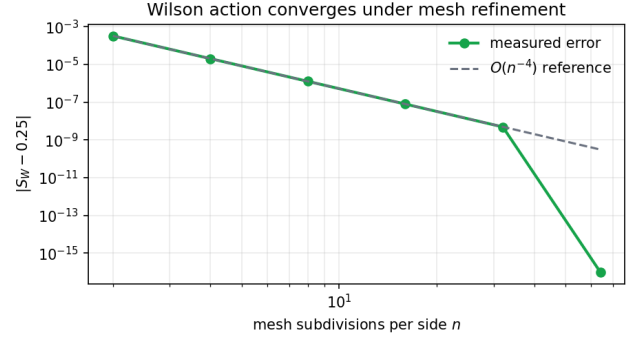
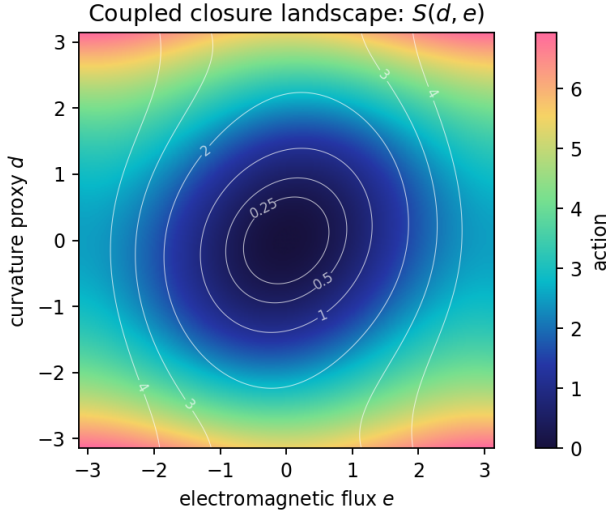


Figure 3: Discrete prototype. **a**, Coupled closure landscape $S(d, e)$; the minimum at $(0, 0)$ is a model property. **b**, Wilson action converges to 0.25 under refinement.

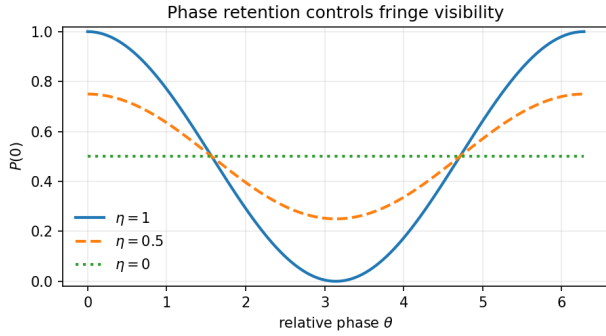


Figure 4: Two-channel phase control. Coherence is imposed; interference verifies the chosen unitary model, not a physical derivation.

adopted for the interference control. The continuous character theorem says that a continuous multiplicative map $A : \mathbb{R} \rightarrow U(1)$ has $A(S) = \exp(i\alpha S)$.

This does not establish unitarity: a finite history kernel K must separately satisfy $K^\dagger K = I$. Nor does it uniquely select complex numbers. For $K = A + iB$,

$$\mathcal{R}(K) = \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \quad (12)$$

is an equivalent paired-real representation, with unitary complex maps becoming orthogonal real maps.

6 Earlier relaxation-labeling audits

The broader framework has three completed empirical software audits.

Signed sparse dynamics. Exact sparse operators matched dense trajectories to floating-point precision across 40 seeds, graph types, mixing values and noise settings. At 100,000 nodes the implementation used 19,999,168 operator bytes; a hypothetical dense signed kernel would require approximately 720 GB.

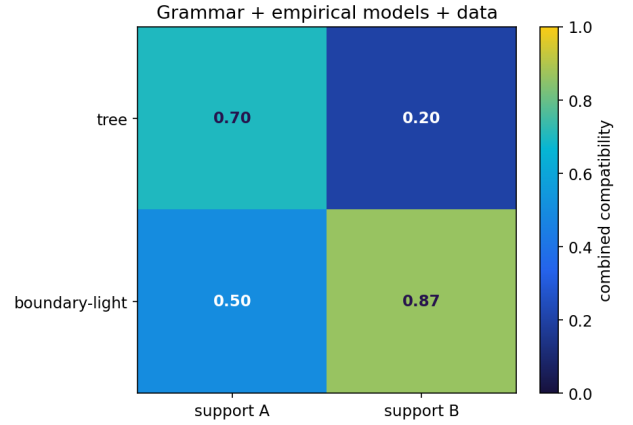


Figure 5: Compatibility is assembled from grammar, empirical-model and data support. The provenance of each source remains inspectable.

Orientation-sensitive compatibility. The cyclic signed chirality factor is exactly decomposable into oriented pair terms. Symmetric undirected pairwise support loses orientation; a hard positive-only predicate retains a genuine higher-order residual.

Causal forecasting. The public forecast example used a batch temporal kernel and full-sample error scale. Causal corrections removed future-data sensitivity. On frozen snapshots, causal prior MAE was 17,872.75 for RSXFSN, 1.2480 for INDPRO and 0.6184 for UNRATE. These results validate the need for causal evaluation, not a universal forecasting advantage.

7 Discussion

The framework’s strongest contribution is architectural. It connects recursive relaxation, focused access to evidence, typed discrete–continuous representation and explicit controls without treating a metaphorical count as a physical dimension. The object–label model supplies a concrete mechanism for the user’s proposed transmo-

grification: attention and information determine role transitions, while the envelope makes external evidence operationally internal to a frame.

Several limitations are decisive. The permutation bridge is synthetic; the current mesh is flat; the matter action is absent; the history measure is not fixed; the Born-form readout is assumed; and no held-out physical observable has been predicted. The Master Observer is an observation map, not a claim about an extra physical observer.

The next experiment should use a non-flat Regge mesh, compact gauge links, matter stress-energy, fixed boundary data and a refinement sequence. Positive-real, signed-real, complex-phase and paired-real models must be compared at matched parameter count. The primary outcomes should be typed residuals, gauge covariance, bridge round-trip error, evolution-commutation defect, unitarity error, held-out observables, runtime and memory. A complex model earns explanatory value only when the bridge gives a held-out restriction or convergence advantage not inserted into its definition.

8 Methods and reproducibility

The source corpus, protocols, synthetic benchmarks and figures are in the relaxation-publication-study directory. The object-label prototype is in `Code/relaxation-labeling/python/grrle/object_label.py`; its four tests pass. The complete study suite passes 17 tests. Figures were generated with deterministic NumPy/Matplotlib scripts and embedded in this PDF. Numerical agreement is agreement with stated mathematics, not agreement with nature.

Data and code availability

Reproducible artifacts include `PROTOCOL.md`, `FORECAST.PROTOCOL.md`, `src/triadic_closure.py`, `src/triangulated_closure.py`, the object-label model, and machine-readable results in `results/`. The publication PDF and source are accompanied by the local study directory.

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